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Data Section Draft

The data in this report comes from Pomona College's Sustainability Office, which surveyed faculty and staff in FY26 to quantify their commuting patterns and habits. The survey was done in line with The South Coast Air Quality Management District (AQMD) Rule 2202, which states that institutions with 250 or more employees within the South Coast Air Basin must measure and reduce air pollution from employee commutes. Since vehicle emissions are a primary source of smog in Southern California, this data is used to enforce Rule 2202 which encourages large employers to reduce the amount of single-occupancy vehicles heading to their worksites.

The data is in the form of two Excel workbooks, with one being for faculty and the other for staff. There is one sheet per fiscal year, originating in FY14. For the purposes of this analysis, only the FY26 Excel sheet is used. The main variables used for analysis are the ZIP code where the respondents report residing, their one-way miles to work, employee type, as well as a mode code for each weekday (Monday-Friday).

Faculty and staff combined together result in 467 total respondents, meaning 32% or 112 of 355 faculty completed the survey, and approximately 43% or 355 of 820 staff responded to the survey. To preserve anonymity, employee IDs are dropped and replaced with an anonymous person_id. General data cleaning practices were conducted such as eliminating trailing spaces and truncating zip codes.

Based on the raw data, the miles which faculty and staff self-report traveling to commute to work range from 1 to 87 miles for faculty and 1 to 60 for staff members. There are 28 faculty ZIP codes and 102 staff ZIP codes. The data was reshaped so that each row would represent one individual person commuting on a given day. In the survey, letter codes were used to describe commuting modality, such as H indicating driving alone, and so on. To interpret these codes, a decoder table was written in the R code so that each of the 29 codes has a mode, category, day type, and whether it counts as an SOV (single-occupancy vehicle). 114 no-trip days were removed if they were recorded as sick or non-commuting days.